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Problem Set 8

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Problem 1: Strong Connectivity

(a) We must prove that R is an equivalence relation. To do this, we need to show that R is reflexive, symmetric, and transitive.

- **Reflexive:** For any vertex $u \in V$, there is a walk of length 0 from u to u. Therefore, a directed walk from u to u exists, meaning uRu holds.
- **Symmetric:** Suppose uRv is true. By definition, there exists a directed walk from u to v and a directed walk from v to u. This logically implies there is a directed walk from v to u and a directed walk from u to v, which is the exact definition of vRu. Thus, $uRv \implies vRu$.
- **Transitive:** Suppose uRv and vRw are both true. This means there are directed walks $u \to v$ and $v \to u$, as well as directed walks $v \to w$ and $w \to v$. By concatenating the walk from $u \to v$ with the walk from $v \to w$, we obtain a valid directed walk from $u \to w$. Similarly, concatenating the walk from $w \to v$ with $v \to u$ yields a directed walk from $w \to u$. Because there is a walk from $u \to w$ and $w \to u$, we have uRw.

Since R satisfies all three properties, it is an equivalence relation.

(b) The equivalence classes of R are exactly the **strongly connected components (SCCs)** of the graph G. By definition, a strongly connected component is a maximal set of vertices such that for every pair of vertices u, v in the set, there is a directed path from u to v and v to u.

Problem 2: Bijections via Inverses

(a) To prove that f is injective, we must show that if $f(a_1) = f(a_2)$ for any $a_1, a_2 \in A$, then $a_1 = a_2$.

Suppose $f(a_1) = f(a_2)$. Applying the function g to both sides, we get: $g(f(a_1)) = g(f(a_2))$ By our assumption, $g(f(a)) = a$ for all $a \in A$. Applying this to both sides yields: $a_1 = a_2$ Thus, f maps distinct inputs to distinct outputs, making it injective.

(b) To prove that f is surjective, we must show that for every $b \in B$, there exists some $a \in A$ such that $f(a) = b$.

Let b be an arbitrary element in B. Define $a = g(b)$. Because $g : B \to A$ is a total function, a is guaranteed to be an element of A. Now, apply f to this a:

$$f(a) = f(g(b))$$

By our assumption, $f(g(b)) = b$ for every $b \in B$. Therefore:

$$f(a) = b$$

Since we found an $a \in A$ for our arbitrary $b \in B$, f is surjective.

(c) A bijection is defined as a function that is both injective and surjective. Since we proved in (a) that f is injective and in (b) that f is surjective, we can conclude that f is a bijection.

Problem 3: Increasing Correspondence

(a) We are given $A = \{(a, b, c) \in \mathbb{N}^3 \mid a \leq b \leq c \text{ AND } a + b + c = 1000\}$ and $B = \{(x, y, z) \in \mathbb{N}^3 \mid x + 2y + 3z = 1000\}$. We must show that $f((a, b, c)) := (c - b, b - a, a)$ is a well-defined function from A to B.

To do this, we must verify two things for any $(a, b, c) \in A$:

(a) **The output is in $\mathbb{N}^3$:** Since $(a, b, c) \in A$, we know $a \leq b \leq c$ and $a, b, c \in \mathbb{N}$. Therefore, $c - b \geq 0$, $b - a \geq 0$, and $a \geq 0$. Thus, $(c - b, b - a, a)$ is an ordered triple of natural numbers.

(b) **The output satisfies the condition for set B:** Let $(x, y, z) = (c - b, b - a, a)$. We evaluate the expression $x + 2y + 3z$:

$$x + 2y + 3z = (c - b) + 2(b - a) + 3(a)$$

$$x + 2y + 3z = c - b + 2b - 2a + 3a = a + b + c$$

Since $(a, b, c) \in A$, we know $a + b + c = 1000$. Thus, $x + 2y + 3z = 1000$, meaning the output belongs to B. This proves f is well-defined.

(b) We want to find a total function $g : B \to A$ that reverses f. Looking at $f(a, b, c) = (x, y, z) = (c - b, b - a, a)$, we can solve for a, b, c:

- $z = a \implies a = z$
- $y = b - a \implies b = y + a = y + z$
- $x = c - b \implies c = x + b = x + y + z$

Thus, we define $g((x, y, z)) := (z, y + z, x + y + z)$.

Now we prove it is well-defined from B to A:

(a) **Output is in $\mathbb{N}^3$ and weakly increasing:** Since $x, y, z \in \mathbb{N}$, it is clear that $z \geq 0$, $y + z \geq 0$, and $x + y + z \geq 0$. Furthermore, since x and y are non-negative, $z \leq y + z \leq x + y + z$, fulfilling the weakly increasing requirement for set A.

(b) **Output sums to 1000:** We sum the coordinates: $z + (y + z) + (x + y + z) = x + 2y + 3z$. Since $(x, y, z) \in B$, $x + 2y + 3z = 1000$. Thus, the output is in A.

Finally, we verify it is an inverse by checking the identities:

- $g(f((a, b, c))) = g(c - b, b - a, a) = (a, (b - a) + a, (c - b) + (b - a) + a) = (a, b, c)$.

- $f(g((x,y,z))) = f(z, y+z, x+y+z) = ((x+y+z)-(y+z), (y+z)-z, z) = (x,y,z)$.

(c) Because f has a valid two-sided inverse g mapping between B and A, we can invoke the result from Problem 2 to conclude that f is a bijection.

Problem 4: Counting Graphs

(a) For the list $(1, 2, \ldots, n)$ to be a valid topological order, every directed edge (i, j) in the graph must flow from left to right in the sequence; this means we must have $i < j$.

The number of possible edges that satisfy $i < j$ in a set of n vertices is simply the number of ways to choose 2 distinct vertices from n, which is $\binom{n}{2}$.

Because a DAG can be formed by taking *any* subset of these valid forward-pointing edges, and each of the $\binom{n}{2}$ edges can independently either be included or excluded from the graph, the total number of valid DAGs is:

$$2^{\binom{n}{2}}$$

(b) A length-3 path consists of a sequence of 4 distinct vertices (v_1, v_2, v_3, v_4). First, we find the number of edges in G. By the Handshake Lemma, $\sum \deg(v) = 2|E|$. Since all n nodes have degree 4, $4n = 2|E|$, so $|E| = 2n$.

Let's build the path by first selecting its "middle" directed step $v_2 \to v_3$. There are $2n$ undirected edges, meaning there are $4n$ directed edges we can choose for this middle step.

Now, we must choose v_1 (which connects to v_2) and v_4 (which connects to v_3):

- v_2 has a degree of 4. One of its incident edges goes to v_3. This leaves 3 choices for the edge connecting to v_1.
- v_3 has a degree of 4. One of its incident edges goes to v_2. This leaves 3 choices for the edge connecting to v_4.

We must ensure the path does not loop back on itself. By definition, a path cannot revisit nodes, so $v_1 \neq v_3$ and $v_4 \neq v_2$ (handled by distinct edges). Furthermore, $v_1 \neq v_4$ because we are explicitly told the graph has no length-3 cycles (triangles). If $v_1 = v_4$, then (v_1, v_2, v_3) would form a triangle, which is forbidden.

Thus, every independent choice of edges yields a valid, unique path.

$$\text{Total length-3 paths} = (4n) \times 3 \times 3 = 36n$$

Problem 5: Counting

(a) There are 5 vowels and 21 consonants in the alphabet. The problem hints that every vowel appears to the left of a consonant. Because the last letter cannot be a vowel, every vowel must be immediately followed by a consonant.

We can think of this as placing the 21 consonants in a line, which can be arranged in 21! ways. This creates exactly 21 valid slots for the vowels (the slot directly in front of each consonant).

We need to select 5 of these 21 slots to place our 5 vowels, which can be done in $\binom{21}{5}$ ways. Finally, the 5 vowels can be permuted among themselves in 5! ways.

Total ways:

$$\binom{21}{5} \times 21! \times 5!$$

(b) We want at least two consonants immediately following each vowel. Imagine placing the 5 vowels in a line, creating 6 "gaps" between/around them:

$$_V_V_V_V_V_$$

Let $x_1, \ldots, x_6$ be the number of consonants in each gap. The constraints are: $x_1 \geq 0$ $x_2 \geq 2, \ldots, x_6 \geq 2$ (Because every vowel, including the last one, must be followed by at least two consonants). The sum of all consonants is $\sum x_i = 21$.

To find the number of ways to distribute the 21 consonants into these gaps, we subtract the minimum requirements ($0+2+2+2+2+2 = 10$) from 21. We now have exactly 11 "excess" consonants to distribute freely among the 6 gaps. Using the stars and bars formula $\binom{n+k-1}{k-1}$, we get $\binom{11+6-1}{6-1} = \binom{16}{5}$ gap configurations.

For each valid layout, we permute the 21 unique consonants (21!) and the 5 unique vowels (5!).

Total ways:

$$\binom{16}{5} \times 21! \times 5!$$

(c) To pair up $2n$ students, imagine lining them up in some arbitrary order. The first student can be paired with any of the remaining $2n-1$ students. Once paired, they step aside. The next unpaired student can be paired with any of the remaining $2n-3$ students. This continues down to 1.

This yields the double factorial $(2n-1)!!$. We can rewrite this using standard factorials by multiplying the numerator and denominator by the even terms $(2n \cdot (2n-2) \ldots 2) = 2^n \cdot n!$.

Total ways:

$$\frac{(2n)!}{n!2^n}$$

(d) Two n-digit sequences are the same type if they are permutations of each other, meaning we only care about the *frequency* of each digit (0 through 9), not the order.

This is a classic stars and bars problem: we are choosing n total digits from a set of 10 possible options (the digits 0-9), with replacement.

Using the combinations with repetition formula $\binom{n+k-1}{k-1}$, where n is the number of selections and $k = 10$ is the number of categories:

Total types:

$$\binom{n+9}{9}$$